

JKBOSE 2026

CLASS 11TH

STATISTICS

MATHS

LECTURE-2

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TOPICS TO BE COVERED

1

Mean Deviation For Grouped Data

2

Exercise Discussion 13.1

Mean deviation for grouped data

Data can be grouped in the following two ways:

- 1) Discrete frequency Distribution.
- 2) Continuous frequency Distribution.

★ Mean Deviation of Discrete frequency Distribution:

Observation	x_1	x_2	x_3	...	x_n
frequency	f_1	f_2	f_3	...	f_n

Here $f_1 + f_2 + f_3 + \dots + f_n = N$.

$$\text{Then M.D.}(\bar{x}) = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{N}$$

$$\& \text{M.D.}(M) = \frac{\sum_{i=1}^n f_i |x_i - M|}{N}.$$

$$\bar{x} = \frac{\sum_{i=1}^n f_i x_i}{N}$$

EXAMPLE

Find mean deviation about the mean for the following data:

x_i	2	5	6	8	10	12
f_i	2	8	10	7	8	5

(Sol) Here $\bar{x} = \frac{300}{40} = 7.5$.

Also $M.D(\bar{x}) = \frac{92}{40}$.

$\Rightarrow M.D(\bar{x}) = 2.3$

x_i	f_i	$f_i x_i$	$x_i - \bar{x}$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
2	2	4	-5.5	5.5	11
5	8	40	-2.5	2.5	20
6	10	60	-1.5	1.5	15
8	7	56	0.5	0.5	3.5
10	8	80	2.5	2.5	20
12	5	60	4.5	4.5	22.5
	40	300			92

EXAMPLE

$$\left(\frac{T_{\frac{n+1}{2}}}{2} \right) \rightarrow \text{odd} \cdot \frac{1}{2} \left(T_{\frac{n}{2}} + T_{\frac{n}{2}+1} \right)$$

Find mean deviation about the **median** for the following data:

x_i	3	6	9	12	13	15	21	22
f_i	3	4	5	2	4	5	4	3

(Sol) Here $N = 30$, therefore the Median is given by;

$$M = \frac{1}{2} \left(T_{\frac{30}{2}} + T_{\frac{30}{2}+1} \right) = \frac{1}{2} (T_{15} + T_{16})$$

$$\Rightarrow M = \frac{1}{2} (13 + 13) = \frac{2 \times 13}{2}$$

$$\Rightarrow \boxed{M = 13}$$

$$\text{Therefore } M.D(M) = \frac{149}{30}$$

$$\Rightarrow \boxed{M.D(M) = 4.97} \text{ Hence the result.}$$

x_i	f_i	cf	$x_i - M$	$ x_i - M $	$f_i x_i - M $
3	3	3	-10	10	30
6	4	7	-7	7	28
9	5	12	-4	4	20
12	2	14	-1	1	2
13	4	18	0	0	0
15	5	23	2	2	10
21	4	27	8	8	32
22	3	30	9	9	27
	30				149

Continuous Frequency Distribution

- A continuous frequency distribution is a **series** in which the data are classified into different **class-intervals** without gaps along with **their respective frequencies**.

$$M.D(\bar{x}) = \frac{\sum_{i=1}^n f_i |x_i - \bar{x}|}{N}$$

x_i — class mark.
of the i^{th} class.

$$M.D(M) = \sum_{i=1}^n f_i |x_i - M|,$$

where

$$M = l + \frac{\left(\frac{N}{2} - cf\right) \times h}{f}$$

* cf is the cumulative of the class preceding the median class

* f is the frequency of the median class.

* h is the size of the class-interval

* l is the lower limit of the median class.

* A class whose cumulative frequency is equal to $\frac{N}{2}$ or just greater than $\frac{N}{2}$ is known as a median class.

EXAMPLE

Find the mean deviation about the mean for the following data:

Marks obtained	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Number of students	2	3	8	14	8	3	2

(Sol) Here $\bar{x} = \frac{1800}{40} = 45$.

So, $M.D(\bar{x}) = \frac{400}{40} = 10$.

$\Rightarrow M.D(\bar{x}) = 10$.

Class Interval	frequency (f_i)	Class Mark (x_i)	$f_i x_i$	$ x_i - \bar{x} $	$f_i x_i - \bar{x} $
10-20	2	15	30	30	60
20-30	3	25	75	20	60
30-40	8	35	280	10	80
40-50	14	45	630	0	0
50-60	8	55	440	10	80
60-70	3	65	195	20	60
70-80	2	75	150	30	60
	40		1800		400

EXERCISE – 13.1

QUESTION



Find the mean deviation about the mean for the data in Exercises 1 and 2.

4, 7, 8, 9, 10, 12, 13, 17

hp/w

QUESTION

Find the mean deviation about the mean for the data in Exercises 1 and 2.

38, 70, 48, 40, 42, 55, 63, 46, 54, 44

k/w

QUESTION



Find the mean deviation about the mean for the data in Exercises 1 and 2.

13, 17, 16, 14, 11, 13, 10, 16, 11, 18, 12, 17

H/w

QUESTION

Find the mean deviation about the mean for the data in Exercises 1 and 2.

36, 72, 46, 42, 60, 45, 5 45, 53, 46, 51, 49

h/w.

QUESTION

Find the mean deviation about the mean for the data in Exercises 5 and 6.

x_i	5	10	15	20	25
f_i	7	4	6	3	5

H/w

QUESTION

Find the mean deviation about the mean for the data in Exercises 5 and 6.

x_i	10	30	50	70	90
f_i	4	24	28	16	8

H/W

QUESTION

Find the mean deviation about the mean for the data in Exercises 5 and 6.

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

H/W

QUESTION

Find the mean deviation about the mean for the data in Exercises 5 and 6.

x_i	15	21	27	30	35
f_i	3	5	6	7	8

hp/w

QUESTION

Find the mean deviation about the mean for the data in Exercises 9 and 10.

Income per day Rs.	0-100	100-200	200-300	300-400	400-500	500-600	600-700	700-800
Number of persons	4	8	9	10	7	5	4	3

H/W.

QUESTION



Find the mean deviation about the mean for the data in Exercises 9 and 10.

Height in cms	95–105	105–115	115–125	125–135	135–145	145–155
Number of boys	9	13	26	30	12	10

h/w.

Thank
You